Notes: MATH270 (W26) Basic Complex Variables

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1 Preliminaries

Definition 1.1.

- Write

$$\begin{aligned} \mathrm{i} &:= \sqrt{-1}, \\ \mathbb{C} &:= \{z = x + \mathrm{i}z : x, y \in \mathbb{R}\}. \end{aligned}$$

- For a complex number $z = x + \mathrm{i}y$, we write its **real part** and **imaginary part** as $\operatorname{Re} z = x$ and $\operatorname{Im} z = y$ respectively.

- For complex numbers $x + \mathrm{i}y$ and $a + \mathrm{i}b$, we define their product as

$$(x + \mathrm{i}y)(a + \mathrm{i}b) := (xa - yb) + \mathrm{i}(xb + ya).$$

- We define the **complex conjugate** of $z = x + \mathrm{i}y$ as $\overline{z} := x - \mathrm{i}y$. Note that $\overline{zw} = \overline{z}\,\overline{w}$, $\overline{z + w} = \overline{z} + \overline{w}$, $\operatorname{Re} z = \frac{z+\overline{z}}{2}$, and $\operatorname{Im} z = \frac{z-\overline{z}}{2\mathrm{i}}$.

- Define the **norm** on $\mathbb{C}$ as

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2}.$$

  With this norm, $\mathbb{C}$ is a normed vector space over $\mathbb{R}$ of dimension 2. Note that this induces the usual Euclidean metric on $\mathbb{R}^2$.

- Define the **inverse** of $z \neq 0$ as

$$z^{-1} := \frac{\overline{z}}{|z|^2} = \frac{x - \mathrm{i}y}{x^2 + y^2}.$$

  One can check that $zz^{-1} = z^{-1}z = 1$.

Remark 1.2. One can view $\mathbb{C}$ as assigning a notion of multiplication to $\mathbb{R}^2$ with some natural properties. What about $\mathbb{R}^d$ for $d \geq 3$. For $d = 4$, this gives the **quaternions** (similar to complex numbers but non-commutative). For $d = 8$, this gives the **octonions** (non-commutative and non-associative).

1.1 Functions in complex variables

A **polynomial** over $\mathbb{C}$ is of the form

$$f(z) := \sum_{k=0}^{n} a_k z^k, \quad a_k \in \mathbb{C}, \quad a_n \neq 0,$$

where n is the **degree** of f.

Example 1.3 (Not all polynomial in $\mathbb{R}^2$ is a polynomial in $\mathbb{C}$). $f(x + \mathrm{i}y)$. A proof will be given later.

A **power series** over $\mathbb{C}$ is of the form

$$f(z) := \sum_{k=0}^{\infty} a_k z^k, \quad a_k \in \mathbb{C}.$$

Lemma 1.4. *Let $\{a_k\}_{k\geq 0} \subset \mathbb{C}$. Define the **radius of convergence R** of the power series $f(z) := \sum_{k=0}^{\infty} a_k z^k$ as*

$$R := \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}} \in [0, \infty].$$

Then, the power series $f(z)$ converges absolutely for all z with $|z| < R$ and diverges for all z with $|z| > R$.

Example 1.5 (The behavior on the boundary $|z| = R$ is more complicated). Consider $\sum_{k\geq 1} z^k/k$. The radius of convergence is $R := \left(\limsup_{k\to\infty} (1/k)^{1/k}\right)^{-1} = 1$. To see this, note that

$$\log (1/k)^{1/k} = -\frac{\log k}{k} \longrightarrow 0 \text{ as } k \to \infty,$$

since the log function grows slower than any polynomial. This power series diverges when $z = 1$ but converges when $z = -1$.

Proof. Since $\mathbb{C}$ has the same topology as $\mathbb{R}^2$, which is complete, we need only show that the resulting series is Cauchy.

Let $N \geq M \geq 0$ and note that

$$\left|\sum_{n=0}^{N} a_n z^n - \sum_{n=0}^{M} a_n z^n\right| = \left|\sum_{n=M+1}^{N} a_n z^n\right| \leq \left|\sum_{n=M+1}^{N} a_n z^n\right| \leq \sum_{n=M+1}^{N} |a_n| \cdot |z|^n.$$

Now, if $|z| < R$, then the series $\sum_{n\geq 0} |a_n||z|^n$ converges (compare with a geometric series). In particular, the right hand side above can be made arbitrarily small by taking N, M large, so the series is Cauchy and hence converges.

If $|z| > R$, then $|a_n z^n| \not\to 0$ as $n \to \infty$. Thus the series cannot converge. □

1.2 Exponential and Logarithm

Definition 1.6. The **exponential function** is defined as

$$e^z := \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad z \in \mathbb{C}.$$

Remark 1.7. $R = \infty$.

Lemma 1.8. *For all $z, w \in \mathbb{C}$, we have $e^{z+w} = e^z e^w$.*

Proof.

$$e^{z+w} = \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{1}{n!} \binom{n}{k} z^k w^{n-k} = \sum_{m\geq 0} \sum_{\ell \geq 0} \frac{1}{m!} \frac{1}{\ell!} z^m w^\ell = e^z e^w,$$

where we use the binomial formula in the second equality. For the third equality, we note that the coefficient of $z^m w^\ell$ on the left hand side is $\frac{1}{(m+\ell)!} \frac{(m+\ell)!}{m!\ell!} = \frac{1}{m!\ell!}$. □

Remark 1.9.

$$e^{\mathrm{i}y} = \sum_{n=0}^{\infty} \frac{\mathrm{i}^n y^n}{n!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} y^{2k} + \mathrm{i} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} y^{2k+1} = \cos y + \mathrm{i} \sin y.$$

☕

Combining this observation with the lemma above, we get the

Lemma 1.10.

$$e^{x+\mathrm{i}y} = e^x(\cos y + \mathrm{i} \sin y).$$

In particular, $|e^{x+\mathrm{i}y}| = e^x$. *As* y *varies,* $e^{x+\mathrm{i}y}$ *spirals around a circle of radius* e^x *with period* 2π.

If $z \in \mathbb{C} \setminus \{0\}$, then $|z/|z|| = 1$. There exists a unique $\theta \in [0, 2\pi)$ such that $e^{\mathrm{i}\theta} = z/|z|$. We call this θ the **argument** of z, denoted $\arg z$. Note that $\arg z$ is not continuous: if we approach the positive real axis from above, $\arg z \to 0$, but if we approach from below, $\arg z \to 2\pi$.

Definition 1.11. Define the **complex logarithm** as

$$\operatorname{Log} z := \log|z| + \mathrm{i} \arg z, \quad z \in \mathbb{C} \setminus \{0\}.$$

Note that $e^{\operatorname{Log} z} = z$ for all $z \in \mathbb{C} \setminus \{0\}$.

This is *not* a canonical logarithm. For each $y \in \mathbb{R}$, we could define $\arg z$ to take values in $[y, y + 2\pi)$ instead of $[0, 2\pi)$. The function $\operatorname{Log}(z)$ with a different choice of y is called a **branch** of $\operatorname{Log}(z)$; it is discontinuous along the ray $\mathbb{R}_{\geq 0} e^{\mathrm{i}y}$. The definition provided above, with $y = 0$, is the **standard branch**.

Other branches of the Log can be useful when, for instance, we want to define $\operatorname{Log} f$ for a function f whose domain includes the positive real axis (but not some other ray).

Definition 1.12. For any $\alpha \in \mathbb{C}$ and any $z \in \mathbb{C} \setminus \{0\}$, we define

$$z^\alpha := e^{\alpha \operatorname{Log} z}.$$

Note that if $\alpha \notin \mathbb{Z}$, this is not continuous or canonically defined. If $\alpha \in \mathbb{Z}$, discontinuities of $\alpha \operatorname{Log} z$ will be multiples of $2\pi\mathrm{i}$, so $e^{\alpha \operatorname{Log} z}$ is continuous and well-defined.

2 Holomorphic Functions

Note that complex multiplication can be viewed as rotation + scaling: for $z = re^{\mathrm{i}\theta}$ and $\tilde{z} = \tilde{r}e^{\mathrm{i}\tilde{\theta}}$, we have

$$z\tilde{z} = r\tilde{r}e^{\mathrm{i}(\theta+\tilde{\theta})}.$$

Definition 2.1. Let $U \subset \mathbb{C}$ be open. A continuous function $f : U \to \mathbb{C}$ is **holomorphic (or complex differentiable) at $z \in U$** if

$$f'(z) = \partial_z f(z) := \lim_{w \to z} \frac{f(w) - f(z)}{w - z}$$

exists. We say f is **holomorphic** if it is holomorphic at each $z \in U$.

Example 2.2. Consider $f(z) := \overline{z}$. Note that

$$\frac{f(w) - f(z)}{w - z} = \frac{\overline{w - z}}{w - z}$$

does not converge as $w \to z$ (the limit is 1 when approached in the $\mathbb{R}$ direction and -1 when approached in the $\mathrm{i}\mathbb{R}$ direction).

Lemma 2.3. *Let $f, g : U \to \mathbb{C}$ be holomorphic. We have the following:*

- *Sum rule: $f + g$ is holomorphic, $(f + g)' = f' + g'$.*
- *Product rule: fg is holomorphic, $(fg)' = f'g + fg'$.*

In addition, we have the

- *Chain rule: if $f : U \to V$, $g : V \to \mathbb{C}$ are holomorphic, then $g \circ f$ is holomorphic, $(g \circ f)' = (g' \circ f)f'$.*

Proof. Same as for real-valued functions. □

Corollary 2.4 (Polynomials are holomorphic)**.** *If $f(z) = \sum_{k=0}^{n} a_k z^k$ is a polynomial in z, then f is holomorphic with complex derivative*

$$f'(z) = \sum_{k=1}^{n} k a_k z^{k-1}.$$

Lemma 2.5 (Power series are holomorphic; term-by-term differentiation)**.** *Let $\{a_k\}_{k \geq 0} \subset \mathbb{C}$. Recall $R := \left(\limsup_{n\to\infty} |a_n|^{1/n}\right)^{-1}$. Then, $f(z) = \sum_{n \geq 0} a_n z^n$ is holomorphic for $|z| < R$ and*

$$f'(z) = \sum_{n \geq 1} n a_n z^{n-1}.$$

Proof. Let $f(z) := \sum_{k=0}^{n} a_k z^k$ denote the partial sums and write $g(z) = \sum_{k \geq 1} k a_k z^{k-1}$. We will show $f' = g$. Note first that since $k^{1/k} \to 1$, g has the same radius of convergence as f.

Fix $r \in (0, R)$. We have when $|z+h| \le r$ that

$$\begin{aligned}
&\left|\frac{f(z+h)-f(z)}{h} - g(z)\right| \\
&\le \left|\frac{f_n(z+h)-f_n(z)}{h} - f_n'(z)\right| + \left|f_n'(z) - g(z)\right| + \left|\frac{(f-f_n)(z+h)-(f-f_n)(z)}{h}\right| \\
&=: A + B + C.
\end{aligned}$$

We first bound C:

$$C = \left|\sum_{k\ge n+1} a_k \frac{(z+h)^k - z^k}{h}\right| \le \sum_{k\ge n+1} |a_k| \cdot \left|\frac{(z+h)^k - z^k}{h}\right|.$$

Using the fact that $a^k - b^k = (a-b)(a^{k-1} + a^{k-2}b + \cdots + b^{k-1})$ and $|z|, |z+h| \le r$, we have

$$\left|\frac{(z+h)^k - z^k}{h}\right| \le kr^{k-1}$$

and so

$$C \le \sum_{k\ge n+1} |a_k| k r^{k-1},$$

which goes to 0 as $n \to \infty$ since $r < R$.

Now, for each $\varepsilon > 0$, there exists N such that

$$\sum_{k\ge n+1} |a_k| k r^{k-1} \le \varepsilon, \quad \forall n \ge N.$$

Since $f_n'(z) \to g(z)$, we can arrange $B \le \varepsilon$ for $n \ge N$. There exists $\delta > 0$ such that with $n \ge N$, $A \le \varepsilon$ when $|h| \le \delta$. Then we can bound

$$A + B + C \le 3\varepsilon, \quad \forall |h| \le \delta.$$

□

Example 2.6. The complex exponential function e^z is holomorphic with derivative $(e^z)' = e^z$.

Corollary 2.7. *Complex power series admits complex derivatives of all orders on its disk of convergence. Moreover,*

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

Proof. The first part follows from repeated application of the previous lemma, which also gives $f^{(n)}(0) = n! a_n$. □

Definition 2.8. A function $f : U \to \mathbb{C}$ is **complex analytic** if for each $z_0 \in U$ there exists $r = r(z_0) > 0$ and $\{a_k\}_{k\ge 0} \subset \mathbb{C}$ (depending on z_0) such that

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k, \quad \forall z \in B_r(z_0) \subset U.$$

By the previous corollary, complex analytic implies infinite complex differentiability.

2.1 The Cauchy-Riemann Equations

Let $U \subset \mathbb{R}^2$ be open and $f : U \to \mathbb{R}^2$ be **real differentiable** at $z = (x, y) \in U$ if there exists a 2×2 matrix $\mathrm{D}f(z)$ (the **Jacobian matrix**) such that

$$f(z + h) = f(z) + \mathrm{D}f(z)h + o(|h|) \text{ as } \mathbb{R}^2 \ni h \to 0.$$

The notation $o(|h|)$ means $\lim_{h\to 0} o(|h|)/|h| = 0$.

If the partial derivatives $\partial_x f(z)$ and $\partial_y f(z)$ exist and are continuous at z, then f is real differentiable at z.

A function $f : \mathbb{C} \supset U \to \mathbb{C}$ can be viewed as a function $f : U \subset \mathbb{R}^2 \to \mathbb{R}^2$ by identifying $x + \mathrm{i}y \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$. We can define thus the partial derivatives of f in the ordinary way.

Proposition 2.9. *Let $U \subset \mathbb{C}$ be open and let $f : u + \mathrm{i}v : U \to \mathbb{C}$ be continuous, where $u := \operatorname{Re} f$ and $v := \operatorname{Im} f$. Then f is **holomorphic at $z \in U$** if and only if f is real differentiable at z and satisfy the **Cauchy-Riemann equations**:*

$$\partial_x u(z) = \partial_y v(z), \quad \partial_y u(z) = -\partial_x v(z).$$

Example 2.10.

- The function $f(z) = \operatorname{Re} z$ is real differentiable. We have $\partial_x u = 1 \neq 0 = \partial_y v$, so f is not holomorphic.
- The function $f(z) = \overline{z}$ has $u = \operatorname{Re} z$ and $v = -\operatorname{Im} z$. Thus, $\partial_x u = 1 \neq -1 = \partial_y v$, so f is not holomorphic.

Proof. By definition, f is holomorphic at z if and only if there exists $f'(z) \in \mathbb{C}$ such that

$$f(z + h) = f(z) + f'(z)h + o(|h|) \text{ as } \mathbb{C} \ni h \to 0.$$

By definition of real differentiability, this is equivalent to f is real differentiable and $\mathrm{D}f(z)$ is the linear map given by multiplication by $f'(z)$ (where we identify $a + \mathrm{i}b \cong (a, b)^\mathsf{T}$). Now recall

$$\mathrm{D}f(z)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \partial_x u \cdot a + \partial_y u \cdot b \\ \partial_x v \cdot a + \partial_y v \cdot b \end{pmatrix}$$

and note that with $f'(z) = s + \mathrm{i}t$, we have

$$f'(z)(a + \mathrm{i}b) = (sa - tb) + \mathrm{i}(ta + sb) \cong \begin{pmatrix} sa - tb \\ ta + sb \end{pmatrix}.$$

Thus, f being holomorphic at z is equivalent to

$$\partial_x u = s, \quad \partial_y u = -t, \quad \partial_x v = t, \quad \partial_y v = s$$

for some $s, t \in \mathbb{R}$. □

Remark 2.11. The proof above shows also that if f is holomorphic at z, then

$$\begin{aligned}\partial_x u(z) = \partial_y v(z) = s = \operatorname{Re} f'(z),\\ -\partial_y u(z) = \partial_x v(z) = t = \operatorname{Im} f'(z),\end{aligned}$$

and

$$\mathrm{D}f(z) = \begin{pmatrix} s & -t \\ t & s \end{pmatrix}.$$

is a rotation + scaling matrix. ☕

Definition 2.12. Let $U \subset \mathbb{R}^2$ be open. A real differentiable function $f : U \to \mathbb{R}^2$ is **conformal** if its Jacobian matrix $\mathrm{D}f(z)$ is the composition of rotation + (non-zero) scaling for each $z \in U$.

Informally, a function is conformal if it is locally angle-preserving. By the Cauchy-Riemann equations, being conformal is equivalent to being holomorphic with non-zero derivative.

Remark 2.13. For $d \geq 3$, every conformal mapping on $U \subset \mathbb{R}^d$ is the composition of rotation, scaling, translation, and inversion ($x \mapsto x/|x|^2$). ☕

2.2 Biholomorphic Functions

Definition 2.14. For $U, V \subset \mathbb{C}$ open, a holomorphic function $f : U \to V$ is a **biholomorphism** if f is a bijection and f^{-1} is holomorphic.

If there exists a biholomorphic function $f : U \to V$, we say U and V are **biholomorphically (or conformally) equivalent**.

Remark 2.15. Some literature uses "conformal map" to mean biholomorphism. ☕

Lemma 2.16. *If $f : U \to \mathbb{C}$ is holomorphic, f' is continuous, and $f'(z) \neq 0$ ($z \in U$), then there exists a neighborhood W of z and neighborhood V of $f(z)$ such that $f|_W : W \to V$ is a biholomorphism.*

Remark 2.17. It turns out that if f is a holomorphic bijection, then automatically $f'(z) \neq 0$ for all $z \in U$. Moreover, the derivative of a holomorphic function is always continuous. These statements will be proved later.

Combining these with the lemma above, we see that the condition that f^{-1} is holomorphic is redundant in the definition of biholomorphism. ☕

Proof. By the proof o the Cauchy-Riemann equations,

$$\mathrm{D}f(z) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad f'(z) = a + \mathrm{i}b.$$

If $f'(z) \neq 0$, then either $a \neq 0$ or $b \neq 0$. In either case, $\det \mathrm{D}f(z) = a^2 + b^2 > 0$. By the inverse function theorem, there exists a neighborhood W of z and a neighborhood V of $f(z)$ such that $f|_W$ is a diffeomorphism from W to V. We check that the inverse

diffeomorphism $f^{-1} : V \to W$ is holomorphic. To do this, we look at $\mathrm{D}f^{-1}(w)$ for $w \in V$. Since $f \circ f^{-1} = \mathrm{Id}$, we know that $\mathrm{D}f\left(f^{-1}(w)\right)\mathrm{D}f^{-1}(w) = I$, which gives

$$\mathrm{D}f^{-1}(w) = \left[\mathrm{D}f\left(f^{-1}(w)\right)\right]^{-1}.$$

This matrix acts on $\mathbb{R}^2$ by multiplication by $1/f'(f^{-1}(w))$. Thus the Cauchy-Riemann equations imply that f^{-1} is holomorphic with

$$(f^{-1})' = \frac{1}{f' \circ f^{-1}}.$$

□

Corollary 2.18. *The standard branch of the logarithm* $\mathrm{Log}(z)$ *is holomorphic on* $\mathbb{C} \setminus [0, \infty)$.

Proof. $\mathrm{Log}(z)$ is the inverse of $z \mapsto e^z$ restricted to the domain $\mathbb{R} \times (0, 2\pi)$, which has non-zero derivative. □

Definition 2.19. A **fractional linear (Mobius) transformation** (FLT) is a function of the form

$$f(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0.$$

The condition $ad - bc \neq 0$ ensures that f is not constant.

Note the following:

(i) Scaling a, b, c, d by $\lambda \in \mathbb{C} \setminus \{0\}$ gives the same function.

(ii) f is defined on $bC \setminus \{-d/c\}$, where $-d/c$ is not defined when $c = 0$ (in which case f is defined everywhere).

(iii) Write $\mathbb{C}_\infty := \mathbb{C} \cup \{\infty\}$. Note that $\mathbb{C}_\infty$ has the same topology as the Riemann sphere S^2 via the stereographic projection. We can view $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$, where we define

$$f\left(-\frac{d}{c}\right) = \infty, \quad f(\infty) = \lim_{z\to\infty} f(z) = \frac{a}{c}.$$

A state a few facts about FLTs:

Lemma 2.20. *If f and $\tilde{f}$ are FLTs with coefficients a, b, c, d and $\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}$, then the composition $f \circ \tilde{f}$ is also a fractional linear transformation with coefficients given by entries of the matrix product*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} \tilde{a} & \tilde{b} \\ \tilde{c} & \tilde{d} \end{pmatrix}.$$

Proof. Direct computation. □

FLTs form a group under composition. This group is isomorphic to the group of 2×2 complex matrices with non-zero determinant, modulo scaling by non-zero complex numbers.

Corollary 2.21. *A FLT is a biholomorphism from* $\mathbb{C} \setminus \{-d/c\}$ *to* $\mathbb{C} \setminus \{a/c\}$.

Proof. If f has coefficients a, b, c, d, then by the precious lemma, the FLT with coefficient matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is the inverse of f. We have $ad - bc \neq 0$ by definition of FLT. Since FLTs are holomorphic where defined, the result follows. □

Lemma 2.22. *If f is a FLT and $A \subset \mathbb{C}$ is a circle or a line, then $f(A)$ is also a circle or a line. Note that a line can be viewed as a circle with infinite radius (joined at ∞).*

Proof. If $c = 0$, then f is a complex affine function, which is a composition of scaling + rotation with translation, which clearly maps circles/lines to circles/lines.

If $c \neq 0$, then

$$f(z) = \frac{ad - bc}{c} \cdot \frac{1}{cz + d} + \frac{a}{c}.$$

That is,

$$f = \text{complex affine} \circ \left(z \mapsto \frac{1}{z} \right) \circ \text{complex affine}.$$

Thus it suffices to prove the result for $f(z) = 1/z$. First consider the circle $C = \{z \in \mathbb{C} : |z - z_0|^2 = r^2\}$. If $w = 1/z \in f(C)$, then

$$r^2 = \left| \frac{1}{w} - z_0 \right|^2 = \left(\frac{1}{w} - z_0 \right) \left(\frac{1}{\overline{w}} - \overline{z}_0 \right) = \frac{1}{w\overline{w}} - \frac{z_0}{\overline{w}} - \frac{\overline{z}_0}{w} + |z_0|^2.$$

Multiply both sides by $|w|^2$, rearrange, and we get

$$\left(r^2 - |z_0|^2 \right) w\overline{w} + z_0 w + \overline{z}_0 \overline{w} = 1.$$

One of the following two cases hold:

- If $r^2 - |z_0|^2 = 0$, we get $z_0\overline{w} + \overline{z}_0\overline{w} = 1$. This is a linear equation in w, which describes a line.
- Otherwise, we rearrange to get

  $$\left(w - z_0' \right) \left(\overline{w} - \overline{z}_0' \right) = \frac{1}{r^2 - |z_0|^2}, \quad z_0' := \frac{z_0}{r^2 - |z_0|^2}.$$

  Since the left hand side is precisely $|w - z_0'|^2$, this describes a circle centered at z_0' with radius $|r^2 - |z_0|^2|^{-1/2}$.

Similarly, we can show that the inversion $z \mapsto 1/z$ maps lines to circles or lines. □

Lemma 2.23. *Let*

$$f(z) := \frac{z - \mathrm{i}}{z + \mathrm{i}}.$$

Then f is a biholomorphism from $H := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ to $B_1(0) := \{z \in \mathbb{C} : |z| < 1\}$ with inverse

$$f^{-1}(z) = \frac{\mathrm{i}(1 + z)}{1 - z}.$$

Proof. We first show f maps the real axis to the unit circle. Let $z = x \in \mathbb{R}$. We have

$$|f(x)| = \frac{|x - \mathrm{i}|}{|x + \mathrm{i}|} = \frac{\sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} = 1.$$

Thus $f(\mathbb{R}) \subset \partial B_1(0)$. Since FLTs maps circles/lines to circles/lines, it must map the line $\mathbb{R} \cup \{\infty\}$ to the circle $\partial B_1(0)$. Thus f must map the upper and lower half planes to disjoint open connected sets whose union is $\mathbb{C} \setminus \partial B_1(0)$. This gives $\{f(H), f(-H)\} = \{B_1(0), \mathbb{C} \setminus \overline{B_1(0)}\}$. There are then only two possibilities. To determine which, note that $f(\mathrm{i}) = 0$. The formula for f^{-1} can be checked by direct computation. □

We have seen the following explicit biholomorphisms:

- $z \mapsto az + b$, $a \in \mathbb{C} \setminus \{0\}$, $b \in \mathbb{C}$: rotation + scaling + translation.
- $z \mapsto (z - \mathrm{i})/(z + \mathrm{i})$: upper half plane to unit disk.
- $\alpha, \beta > 0$, $\alpha\beta \leq 2\pi$. Then $z \mapsto z^\alpha$ is a biholomorphism from the wedge $\{z \in \mathbb{C} : 0 < \arg z < \beta\}$ to the wedge $\{z \in \mathbb{C} : 0 < \arg z < \alpha\beta\}$.
- $z \mapsto e^z$. For each $\alpha \in (0, 2\pi)$, this function maps the strip $\{z \in \mathbb{C} : 0 < \operatorname{Im} z < \alpha\}$ to the wedge $\{z \in \mathbb{C} : 0 < \arg z < \alpha\}$. It has inverse $z \mapsto \operatorname{Log} z$.

By composing these maps, we can get many more explicit biholomorphisms. In fact, we will later see that any two simply connected proper open subsets of $\mathbb{C}$ are biholomorphically equivalent (Riemann mapping theorem).

Example 2.24. Find a biholomorphism from the lower half disk $U := \{z \in \mathbb{C} : |z| < 1, \operatorname{Im} z < 0\}$ to the half plane $H := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$.

We first map U to a wedge. Let $g(z) := \mathrm{i}(1 + z)/(1 - z)$. We have seen that g maps the unit disk to the upper half plane. What is $g(U)$?

Note that $g(0) = \mathrm{i}$, $g(-1) = 0$, and $g(1) = \infty$. Thus $g(\mathbb{R})$ is a circle or line which includes $\mathrm{i}, 0, \infty$, the only possibility is the imaginary axis, $g(\mathbb{R}) = \mathrm{i}\mathbb{R}$. Thus also $g((-1, 1)) = \mathrm{i}(0, \infty)$. By the same topological argument as before, either $g(U) = \{z \in H : \operatorname{Re} z > 0\}$ or $g(U) = \{z \in H : \operatorname{Re} z < 0\}$. To determine which, note that $g(-\mathrm{i}) = 1$. Thus,

$$g(U) = \{z \in H : \operatorname{Re} z > 0\} = \left\{z \in \mathbb{C} : 0 < \arg z < \frac{\pi}{2}\right\}.$$

This is the wedge with angle $\pi/2$, which can be mapped to H by the biholomorphism $z \mapsto z^2$. The desired biholomorphism is thus

$$f(z) = \left(\frac{\mathrm{i}(1 + z)}{1 - z}\right)^2.$$

3 Complex Line Integrals

Definition 3.1. Let $U \subset \mathbb{C}$. A **curve** in U is a continuously differentiable function $\gamma : [a, b] \to U$ for some $a < b$. We say γ is **simple** if it is injective. We say γ is a **loop** if $\gamma(a) = \gamma(b)$.

Remark 3.2. A curve with finitely many "kinks" admits a continuous differentiable parameterization. To do this, we "slow down" the parameterization near the kinks so that the derivative there is 0.

Definition 3.3. Let $U \subset \mathbb{C}$, $\gamma : [a, b] \to U$ a curve, and $f : U \to \mathbb{C}$. The **complex line integral** of f over γ is

$$\oint_\gamma f(z)\,\mathrm{d}z := \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t,$$

where the derivative of $\gamma(t) = \gamma_1(t) + \mathrm{i}\gamma_2(t)$ is defined as

$$\gamma'(t) := \gamma_1'(t) + \mathrm{i}\gamma_2'(t).$$

Example 3.4. Let $\gamma(t) := e^{\mathrm{i}t}$ for $t \in [0, 2\pi]$. Let $n \in \mathbb{Z}$.

$$\oint_\gamma z^n\,\mathrm{d}z = \int_0^{2\pi} e^{\mathrm{i}nt} \cdot \mathrm{i}e^{\mathrm{i}t}\,\mathrm{d}t.$$

If $n = -1$, we have

$$\int_0^{2\pi} \mathrm{i}\,\mathrm{d}t = 2\pi\mathrm{i}.$$

If $n \neq -1$, we have

$$\int_0^{2\pi} e^{\mathrm{i}(n+1)t}\mathrm{i}\,\mathrm{d}t = \mathrm{i} \cdot \frac{1}{\mathrm{i}(n+1)} e^{\mathrm{i}(n+1)t}\Big|_0^{2\pi} = 0$$

by periodicity. Thus,

$$\oint_\gamma z^n\,\mathrm{d}z = \begin{cases} 2\pi\mathrm{i}, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Lemma 3.5 (Line integral is invariant under reparameterization)**.** *Let $\varphi : [c, d] \to [a, b]$ be increasing and continuous differentiable. Let $\tilde{\gamma} := \gamma \circ \varphi : [c, d] \to U$. Then*

$$\oint_{\tilde{\gamma}} f(z)\,\mathrm{d}z = \oint_\gamma f(z)\,\mathrm{d}z.$$

If instead φ is decreasing, then

$$\oint_{\tilde{\gamma}} f(z)\,\mathrm{d}z = -\oint_\gamma f(z)\,\mathrm{d}z.$$

Proof. With the change of variables $s = \varphi(t)$, we have

$$\begin{aligned}\oint_{\tilde{\gamma}} f(z)\,\mathrm{d}z &= \int_c^d f(\gamma(\varphi(t)))\gamma'(\varphi(t))\varphi'(t)\,\mathrm{d}t \\ &= \int_a^b f(\gamma(s))\gamma'(s)\,\mathrm{d}s = \oint_\gamma f(z)\,\mathrm{d}z.\end{aligned}$$

The case when φ is decreasing is similar, except the limits of integration are swapped. □

Definition 3.6. Let $V \subset U$ be a subdomain such that ∂V is a simple loop in U. Define $\oint_{\partial V} f(z)\,\mathrm{d}z \coloneqq \oint_\gamma f(z)\,\mathrm{d}z$, where γ is a continuously differentiable curve which traverses ∂V counterclockwise.

Definition 3.7. The **length** of a curve $\gamma : [a, b] \to \mathbb{C}$ is $\operatorname{len}\gamma \coloneqq \int_a^b |\gamma'(t)|\,\mathrm{d}t$.

Proposition 3.8. len *is invariant under reparameterization.*

Lemma 3.9.

$$\left|\oint_\gamma f(z)\,\mathrm{d}z\right| \le \sup_{t\in[a,b]} |f(\gamma(t))| \cdot \operatorname{len}\gamma.$$

Proof.

$$\begin{aligned}\left|\oint_\gamma f(z)\,\mathrm{d}z\right| = \left|\int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t\right| &\le \int_a^b |f(\gamma(t))| \cdot |\gamma'(t)|\,\mathrm{d}t \\ &\le \sup_{t\in[a,b]} |f(\gamma(t))| \cdot \int_a^b |\gamma'(t)|\,\mathrm{d}t \\ &= \sup_{t\in[a,b]} |f(\gamma(t))| \cdot \operatorname{len}\gamma.\end{aligned}$$

□

Lemma 3.10. *Let $f : U \to \mathbb{C}$ be a holomorphic function. For any curve $\gamma : [a, b] \to U$, we have*

$$\oint_\gamma f'(z)\,\mathrm{d}z = f(\gamma(b)) - f(\gamma(a)).$$

Thus if γ is a loop, then $\oint_\gamma f'(z)\,\mathrm{d}z = 0$.

Proof.

$$\begin{aligned}\oint_\gamma f'(z)\,\mathrm{d}z = \int_a^b f'(\gamma(t))\gamma'(t)\,\mathrm{d}t &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t))\,\mathrm{d}t \\ &= f(\gamma(b)) - f(\gamma(a)).\end{aligned}$$

□

Definition 3.11. A **triangle** is the closed region bounded by 3 line segments.

Lemma 3.12 (Goursat)**.** *Let $f : U \to \mathbb{C}$ be holomorphic. Let $T \subset U$ be a triangle. Then $\oint_{\partial T} f(z)\,\mathrm{d}z = 0$.*

Proof. Let a, b, c be the vertices of T. Let u, v, w be the midpoints of the edges. Subdivides T into $\Delta_1, \ldots, \Delta_4 \subset T$. Note that $\sum_j \oint_{\partial\Delta_j} f(t)\,\mathrm{d}z = \oint_{\partial T} f(z)\,\mathrm{d}z$ and $\operatorname{len}\Delta_j = \frac{1}{2}\operatorname{len}\partial T$. Thus, there exists j such that

$$\left|\oint_{\partial\Delta_j} f(z)\,\mathrm{d}z\right| \geq \frac{1}{4}\left|\oint_{\partial T} f(z)\,\mathrm{d}z\right|.$$

Let $T_1 \coloneqq \Delta_j$ for this j. We can iterate this process to get a sequence of nested sets $T \supset T_1 \supset T_2 \supset \ldots$ with $\operatorname{len}\partial T_n = 2^{-n}\operatorname{len}\partial T$ and

$$\left|\oint_{\partial T_n} f(z)\,\mathrm{d}z\right| \geq 4^{-n}\left|\oint_{\partial T} f(z)\,\mathrm{d}z\right|.$$

The nested intersection $\bigcap_{n=0}^{\infty} T_n$ is a single point $z_0 \in U$. Since f is holomorphic at z_0, for each $\varepsilon > 0$ there exists $r > 0$ such that

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon|z - z_0|, \quad \forall z \in B_r(z_0).$$

Find large n such that $T_n \subset B_r(z_0)$ and we have

$$\oint_{\partial T_n} (z - z_0)f'(z_0)\,\mathrm{d}z = \oint_{\partial T_n} \partial_z\left[\frac{1}{2}(z - z_0)^2 f'(z_0)\right]\,\mathrm{d}z = 0.$$

Similarly,

$$\oint_{\partial T_n} f(z_0)\,\mathrm{d}z = 0.$$

We now bound $\left|\oint_{\partial T_n} f(z)\,\mathrm{d}z\right|$:

$$\begin{aligned}
\left|\oint_{\partial T_n} f(z)\,\mathrm{d}z\right| &= \left|\oint_{\partial T_n} \left[f(z) - f(z_0) - f'(z_0)(z - z_0)\right]\,\mathrm{d}z\right| \\
&\leq \operatorname{len}(\partial T_n) \cdot \sup_{z\in\partial T_n} |f(z) - f(z_0) - f'(z_0)(z - z_0)| \\
&\leq 2^{-n}\operatorname{len}(\partial T) \cdot \underbrace{\varepsilon \sup_{z\in\partial T_n} |z - z_0|}_{\leq \operatorname{len}\partial T_n} \\
&\leq \varepsilon \cdot 4^{-n}(\operatorname{len}\partial T)^2.
\end{aligned}$$

Thus,

$$4^{-n}\left|\oint_{\partial T} f(z)\,\mathrm{d}z\right| \leq \left|\oint_{\partial T_n} f(z)\,\mathrm{d}z\right| \leq \varepsilon 4^{-n}(\operatorname{len}\partial T)^2.$$

By sending $\varepsilon \to 0$, we have $|\oint_{\partial T} f(z)\,\mathrm{d}z| = 0$. □

Definition 3.13. $U \subset \mathbb{C}$ is **convex** if for any $z, w \in U$, the line segment joining z and w is contained in U.

Lemma 3.14. *If $U \subset \mathbb{C}$ is convex and open, and $f : U \to \mathbb{C}$ is holomorphic, then there exists a holomorphic $F : U \to \mathbb{C}$ such that $F' = f$ on U.*

Corollary 3.15. *The line integral of holomorphic functions along a loop is zero on a convex domain.*

Proof *(of Lemma)***.** Let $z_0 \in U$. Define $F(z) := \oint_{[z_0,z]} f(u)\,\mathrm{d}u = \int_0^1 f(tz + (1-t)z)(z - z_0)\,\mathrm{d}t$. We will show that $F' = f$. For $z, w \in U$, consider the triangle T with boundary $\partial T = [z_0, z] \cup [z, u] \cup [w, z_0]$. By Goursat, $0 = \oint_{\partial T} f(z)\,\mathrm{d}z = F(z) - F(w) + \oint_{[z,w]} f(w)\,\mathrm{d}w$. Thus we have

$$\frac{F(z) - F(w)}{z - w} = \frac{1}{z - w} \oint_{[w,z]} f(w)\,\mathrm{d}w = \int_0^1 f(tw + (1-t)z)\,\mathrm{d}t \to f(z) \text{ as } w \to z.$$

□

4 Cauchy Integral Formula

4.1 Homotopy

Definition 4.1. A **homotopy** from γ_0 to γ_1 is a continuous map $H : [0,1] \times [0,1] \to U$ such that

$$\begin{aligned} H(0,t) &= \gamma_0(t), \\ H(1,t) &= \gamma_1(t), \\ H(s,0) &= \gamma_0(0) = \gamma_1(0), \\ H(s,1) &= \gamma_0(1) = \gamma_1(1). \end{aligned}$$

Note that if $U \subset V$, then H is also a homotopy in V. The curves γ_0, γ_1 are **homotopic** in U if there exists a homotopy from γ_0 to γ_1 in U.

Example 4.2. (i) $\gamma_0(t) = e^{2\pi i t}$ and $\gamma_1(t) = 2e^{2\pi i t} - 1$ are homotopic in $\mathbb{C} \setminus \{0\}$ with $H(s,t) = (1-s)\gamma_0(t) + s\gamma_1(t)$.

(ii) $\gamma_0(t) = e^{2\pi i t}$ and $\gamma_1(t) = 1$ are not homotopic in $\mathbb{C} \setminus \{0\}$.

Definition 4.3. $U \subset \mathbb{C}$ is **simply connected** if every loop in U is homotopic to a constant loop.

Intuitively, this is true when U has no holes.

Example 4.4. If U is convex, then U is simply connected. For any loop $\gamma : [0,1] \to U$ and any $z_0 = \gamma(0) = \gamma_1$, we have $H(s,t) = sz_0 + (1-s)\gamma(t) \in U$ is a homotopy.

Theorem 4.5. *Let $U \subset \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be holomorphic. Let $\gamma_0, \gamma_1 : [0,1] \to U$ be curves with the same endpoints and homotopic via a continuous differentiable homotopy. Then,*

$$\oint_{\gamma_0} f(z)\,\mathrm{d}z = \oint_{\gamma_1} f(z)\,\mathrm{d}z.$$

Proof. Let $H : [0,1]^2 \to U$ be the continuously differentiable homotopy. Then $K := H([0,1]^2) \subset U$ is compact. Each point of $z \in K$ has an $r_z > 0$ such that $B_{r_z}(z) \subset U$. Since K is compact, it has a finite cover. Thus there exists $\varepsilon > 0$ such that $B_\varepsilon(K) \subset U$. By continuity, there exists $\delta > 0$ such that

$$|H(s,t) - H(s',t')| < \frac{\varepsilon}{4}, \quad \forall |(s,t) - (s',t')| < \delta.$$

We choose $n \in \mathbb{N}$ such that $n^{-1} < \delta$. For each $n^{-1} \times n^{-1}$ square S, let n_S be the loop which is the image of ∂S under H. By our choice of ε and δ, $n_S \subset B_\varepsilon(z) \subset U$ for some $z \in K$. In particular, $B_\varepsilon(z)$ is convex. Thus by a previous result, we have $\oint_{n_S} f(z)\,\mathrm{d}z = 0$.

Now, for columns $k = 1, \ldots, n$, we have

$$0 = \sum_{S \subset \text{column } k} \oint_{n_S} f(z)\,\mathrm{d}z = \oint_{H(\frac{k}{n},\cdot)} f(z)\,\mathrm{d}z - \oint_{H(\frac{k-1}{n},\cdot)} f(z)\,\mathrm{d}z.$$

In particular, $\oint_{H(1,\cdot)} f(z)\,\mathrm{d}z = \oint_{H(0,\cdot)} f(z)\,\mathrm{d}z$, which is the desired result. □

Corollary 4.6. *If $U \subset \mathbb{C}$ is simply connected and $f : U \to \mathbb{C}$ is holomorphic, then $\int_\gamma f(z)\,\mathrm{d}z = 0$ for all loops γ in U.*

Proof. Homotopy invariance + complex integral over a constant loop is zero. □

Remark 4.7. The converse of this corollary is also true. For the remainder of the class, it suffices to take this as the definition of simply connected.

Example 4.8. Let γ traverse $\partial[-1,1]$ once in counterclockwise direction, which is homotopic to a loop traversing $\partial B_1(0)$ once in counterclockwise direction in $\mathbb{C} \setminus \{0\}$. We have for each $n \in \mathbb{Z}$

$$\oint_\gamma z^n\,\mathrm{d}z = \oint_{\partial B_1(0)} z^n\,\mathrm{d}z = \begin{cases} 2\pi\mathrm{i}, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

Corollary 4.9. *If $U \subset \mathbb{C}$ is simply connected and $f : U \to \mathbb{C}$ is holomorphic, then there exists $F : U \to \mathbb{C}$ such that $F' = f$.*

Proof. Fix $z_0 \in U$. For $z \in U$, let γ_z be a curve in U from z_0 to z. Define $F(z) := \oint_{\gamma_z} f(u)\,\mathrm{d}u$. Let $\tilde{\gamma}_z$ be a curve in U from z_0 to z. Then $\tilde{z}_z^- * \gamma_z$ is a loop starting and ending at z_0. Thus, $\oint_{\gamma_z} f(u)\,\mathrm{d}u = \oint_{\tilde{\gamma}_z} f(u)\,\mathrm{d}u$, so F is well-defined.

We now want to show $F' = f$. Let $z, w \in U$ with $|z - w|$ small enough such that $[z, w] \subset U$. Consider a loop which traverses γ_z, then $[z, w]$, then γ_w^-, which has integral zero:

$$F(z) - F(w) + \oint_{[z,w]} f(u)\,\mathrm{d}u = 0.$$

Rearrange and send $z \to w$ to get $F'(w) = f(w)$. □

Example 4.10. $f(z) = z^{-1}$ has no holomorphic antiderivative on $\mathbb{C} \setminus \{0\}$.

4.2 Cauchy Integral Formula

Theorem 4.11 (Cauchy Integral Formula)**.** *Let $U \subset \mathbb{C}$ be open, $r > 0$, and $z_0 \in U$ be such that $B_r(z_0) \subset U$. Let $f : U \to \mathbb{C}$ be holomorphic. Then,*

$$f(z) = \frac{1}{2\pi\mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{u - z}\,\mathrm{d}u, \quad \forall z \in B_r(z_0).$$

Proof. Let $\varepsilon > 0$. Consider the holomorphic function $u \mapsto \frac{f(u)}{u-z}$ on $U \setminus \{z\}$ and the curve $\gamma_0(t) = z_0 + re^{2\pi\mathrm{i}t}$ for $t \in [0,1]$, and let γ_1 traverse $[z_0 + r, z + \varepsilon]$ then $\partial B_\varepsilon(z)$

then $[z+\varepsilon, z_0+r]$. Note that γ_0 and γ_1 are homotopic in $U \setminus \{z\}$. We have

$$\begin{aligned}\frac{1}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{u-z}\,\mathrm{d}u &= \frac{1}{2\pi \mathrm{i}} \oint_{\gamma_1} \frac{f(u)}{u-z}\,\mathrm{d}u \\ &= \frac{1}{2\pi \mathrm{i}} \oint_{\partial B_\varepsilon(z)} \frac{f(u)}{u-z}\,\mathrm{d}u \\ &= \frac{1}{2\pi \mathrm{i}} \int_0^{2\pi} \frac{f(\varepsilon e^{\mathrm{i}t}+z)}{\varepsilon e^{\mathrm{i}t}} \varepsilon \mathrm{i} e^{\mathrm{i}t}\,\mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} f(\varepsilon e^{\mathrm{i}t}+z)\,\mathrm{d}t.\end{aligned}$$

As $\varepsilon \to 0$, the last term converges to $f(z)$ by continuity. □

Example 4.12.

$$\oint_{\partial[-3,3]^2} \frac{e^z \cos(z^2)}{z-1}\,\mathrm{d}z = \oint_{\partial B_3(0)} \frac{e^z \cos(z^2)}{z-1}\,\mathrm{d}z = 2\pi \mathrm{i} \cdot e \cos(1).$$

Theorem 4.13. *Let $f : U \to \mathbb{C}$ be holomorphic. Then for each $z_0 \in U$ and $r > 0$ such that $B_r(z_0) \subset U$, there exists a sequence $\{a_n\}_{n\geq 0} \subset \mathbb{C}$ such that*

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n, \quad \forall z \in B_r(z_0),$$

Proof.

$$\begin{aligned}f(z) &= \frac{1}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{u-z_0} \frac{u-z_0}{u-z}\,\mathrm{d}u = \frac{1}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{u-z_0} \frac{1}{1-\frac{z-z_0}{u-z_0}}\,\mathrm{d}u \\ &= \frac{1}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{u-z_0} \sum_{n=0}^{\infty} \left(\frac{z-z_0}{u-z_0}\right)^n \mathrm{d}u = \sum_{n=0}^{\infty} (z-z_0)^n \frac{1}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{(u-z_0)^{n+1}}\,\mathrm{d}u.\end{aligned}$$

□

Corollary 4.14. *Every holomorphic function has complex derivatives of all orders.*

Corollary 4.15 (Cauchy estimate)**.** *From*

$$f^{(n)}(z_0) = n! a_n = \frac{n!}{2\pi \mathrm{i}} \oint_{\partial B_r(z_0)} \frac{f(u)}{(u-z_0)^{n+1}}\,\mathrm{d}u,$$

we have

$$|f^{(n)}(z_0)| \leq \frac{n!}{r^n} \sup_{u \in \partial B_r(z_0)} |f(u)|.$$

Proposition 4.16 (Liouville's Theorem)**.** *Suppose $f : \mathbb{C} \to \mathbb{C}$ is holomorphic and there exists $C > 0$ and $n \in \mathbb{N}$ such that $|f(z)| \leq C(1+|z|)^n$ for $z \in \mathbb{C}$. Then f is a polynomial of degree less than n.*

Proof. By Cauchy's estimate, for any $z_0 \in \mathbb{C}$ and $r > 0$,

$$|f^{(n+1)}(z_0)| \leq \frac{(n+1)!}{r^{n+1}} \sup_{z \in B_r(z_0)} |f(z)| \leq \frac{(n+1)!}{r^{n+1}} C(1 + |z_0| + r)^n.$$

Since the last term goes to 0 as $r \to \infty$, we know that $f^{(n+1)}(z_0) = 0$. Thus $f^{(n)}(z_0)$ is constant and f is a polynomial with degree at most n. □

Corollary 4.17. *If f is bounded and holomorphic, then f is a constant.*

Proof. Take $n = 0$ in the proposition. □

4.3 The Fundamental Theorem of Algebra

Corollary 4.18 ($\mathbb{C}$ is algebraically closed)**.** *Let f be a nonconstant polynomial over $\mathbb{C}$. Then there exists $z \in \mathbb{C}$ such that $f(z) = 0$.*

Proof. If not, then $1/f$ is holomorphic. Since f is a polynomial, $f(z) \to \infty$ as $|z| \to \infty$. Thus $1/f(z) \to 0$ and is in particular bounded. By Liouville's theorem, $1/f$ is constant, a contradiction. □

Corollary 4.19 (Fundamental Theorem of Algebra)**.** *Let f be a nonconstant polynomial over $\mathbb{C}$. Then f has exactly n zeros, counted with multiplicity.*

Proof. By algebraic closeness, there exists some zero c, so $f(z) = (z - c)g(z)$, where $g(z)$ is a polynomial of degree $n - 1$. If $(n - 1) \geq 0$, then g must have its own zero c_2. By induction,

$$f(z) = (z - c_1)(z - c_2) \cdots (z - c_n).$$

□

Theorem 4.20 (Morera)**.** *Let $U \subset \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be continuous. Let τ be a collection of triangles in U which contains all sufficiently small triangles in U. If $\oint_{\partial T} f(z)\, \mathrm{d}z = 0$ for each $T \in \tau$, then f is holomorphic.*

Proof. It suffices to translate and check holomorphicity at 0. Assume wlog that $0 \in U$ and show f is holomorphic at 0.

Take $\varepsilon > 0$ such that $B_\varepsilon(0) \subset U$ and τ contains every triangle containing 0 inside $B_\varepsilon(0)$. Define

$$F(z) := \oint_{[0,z]} f(w)\, \mathrm{d}w = z \int_0^1 f(tz)\, \mathrm{d}t.$$

Note that we have

$$\begin{aligned} F(z') - F(z) &= \int_{[0,z']} f(w)\, \mathrm{d}w - \int_{[0,z]} f(w)\, \mathrm{d}w \\ &= -\oint_{\partial T} f(w)\, \mathrm{d}w + \int_{[z,z']} f(w)\, \mathrm{d}w = (z' - z) \int_0^1 f(z + t(z' - z))\, \mathrm{d}t, \end{aligned}$$

where T is the triangle with vertices $0, z, z'$. Thus,

$$\frac{F(z') - F(z)}{z' - z} = \int_0^1 f(z + t(z' - z))\, \mathrm{d}t.$$

Sending $z' \to z$ and we get $F'(z)$ exists and equals $f(z)$ on $B_\varepsilon(0)$. Thus we have that f is the derivative of a holomorphic function and thus is also holomorphic. □

Lemma 4.21. *Let U be open and simply connected and f be a holomorphic function on U. If $f(z) \neq 0$ on U, then there exists $g : U \to \mathbb{C}$ such that g is holomorphic and $e^g = f$. Moreover, for each $n \in \mathbb{Z} \setminus \{0\}$, there exists holomorphic $h_n : U \to \mathbb{C}$ such that $(h_n)^n = f$ (take $h_n = e^{g/n}$).*

Proof. Since f'/f is holomorphic, there exists a holomorphic g such that $g = f'/f$. Then $(fe^{-g})' = 0$, so e^g is a nonzero constant multiple of f, say $e^g = cf$ for some $c \in \mathbb{C} \setminus \{0\}$. By replacing g by $g - \log(c)$, we have $e^g = f$. □

5 Open Mapping Theorem

Definition 5.1. A topological space U is **connected** if the only sets that are closed and open are $\varnothing$ and U. Equivalently, U is connected if it cannot be written as $A \cup B$ for nonempty disjoint open sets A, B.

Lemma 5.2. *If U is open and connected, $f : U \to \mathbb{C}$ is holomorphic, and there exists $z \in U$ such that $f^{(n)}(z) = 0$ for all $n \geq 1$, then f is constant.*

Proof. Define $A = \{z \in U : f^{(n)}(z) = 0, \forall n \geq 1\}$. We claim that A is closed and open. To see that A is closed, note that $A_n = \{z \in U : f^{(n)}(0)\}$ is closed and $A = \cap_n A_n$. To see that A is open, note that for each $z \in A$, we can express f in a neighborhood around z as a power series with zero coefficients. Since $z \in A$ and U is connected, we have $A = U$, so f is constant. □

Definition 5.3. Let $f : U \to \mathbb{C}$ and $z \in U$ be a zero of f. The **order** of z_0 is the number N satisfying $N = \min\{n \geq 1 : f^{(n)}(z_0) \neq 0\}$.

Lemma 5.4. *Let $f : U \to \mathbb{C}$ be holomorphic and nonconstant. Let $z_0 \in U$ be a zero of f and N its order. There exists a neighborhood V of z_0 such that $f(z) = (z - z_0)^N h(z)$ on V, where h is a function holomorphic on V and $h(z_0) \neq 0$.*

Proof. Coefficients of power series for f satisfy $a_n = f^{(n)}(z_0)/n!$, so

$$f(z) = \sum_{n=N}^{\infty} a_n (z - z_0)^n = (z - z_0)^N h(z),$$

where $h(z)$ is also holomorphic with $a_N = h(z_0) \neq 0$, since $a_N = f^{(n)}(z_0)/n! \neq 0$. □

Definition 5.5. For $A \subset \mathbb{C}$, $z \in \mathbb{C}$ is an **accumulation point** of A if for every $\varepsilon > 0$, we have $A \cap (B_\varepsilon(z) \setminus \{z\}) \neq \varnothing$.

Lemma 5.6. *If U is open and connected, $f : U \to \mathbb{C}$ holomorphic, and the set $\{z \in U : f(z) = 0\}$ has an accumulation point, then f is constant.*

Proof. Assume that f is not constant. If z_0 is an accumulation point of $\{f = 0\}$, then $f(z_0) = 0$ by continuity. Fix $z_0 \in U$ such that $f(z_0) = 0$. We will show that the zeros of f do not accumulate at $z_0 = 0$ and thus $\{f = 0\}$ has no accumulation points.

Let N be the order of z_0, so $f(z) = (z - z_0)^N h(z)$ in a neighborhood of z with h holomorphic and $h(z_0) \neq 0$. This implies that there exists $\varepsilon > 0$ such that for each $z \in B_\varepsilon(z_0)$, $h(z) \neq 0$. Thus, $f \neq 0$ on $B_\varepsilon(z_0) \setminus \{z_0\}$ and so z_0 cannot be an accumulation point of $\{f = 0\}$. □

Proposition 5.7 (Unique Continuation for Holomorphic Functions)**.** *Let $U \subset \mathbb{C}$ be open and connected. Let f, g be holomorphic on U. If the set of points $\{z \in U : f(z) = g(z)\}$ has an accumulation point, then $f = g$.*

Proof. Apply preceding lemma to $f - g$. □

Theorem 5.8 (Opening Mapping)**.** *Let $U \subset \mathbb{C}$ be open, connected, and $f : U \to \mathbb{C}$ be holomorphic and nonconstant. Then f maps open sets to open sets.*

Proof. Fix $z_0 \in U$. We will show that for every $\varepsilon > 0$ such that $B_\varepsilon(z_0) \subset U$, the set $f(B_\varepsilon(z_0))$ contains a neighborhood of $f(z_0)$. Consider the following cases:

- $f'(z_0) \neq 0$. By the Cauchy-Riemann equations, $\mathrm{D}f(z_0)$ is invertible. The inverse function theorem states that f is a diffeomorphism from a neighborhood of z_0 to a neighborhood of $f(z_0)$.

- $f'(z_0) = 0$. Let N be the order of $f(z) = f(z_0)$, so $N \geq 2$. In a neighborhood of z_0,
$$f(z) - f(z_0) = (z - z_0)^N h(z),$$
where h is holomorphic and $h(z_0) \neq 0$. Thus there exists $\delta > 0$ such that $h(z) \neq 0$ for all $z \in B_\delta(z_0)$. By the existence of a holomorphic log, there exists $g : B_\delta(0) \to \mathbb{C}$ holomorphic such that $h = g^N$. Then,
$$f(z) - f(z_0) = [(z - z_0)g(z)]^N, \quad \forall z \in B_\delta(z_0).$$
By the product rule, the derivative of $z \mapsto (z - z_0)g(z)$ is nonzero at z_0, so by the inverse function theorem, the image under $z \mapsto (z - z_0)g(z)$ of every neighborhood of z_0 contains a neighborhood of 0. Further, $f(z) = [(z - z_0)g(z)]^N + f(z_0)$. We can show that $w \mapsto f(z_0) + w^n$ is an open mapping, so the composition $f(z) = [(z - z_0)g(z)]^N + f(z_0)$ is an open mapping.

□

Corollary 5.9. *Let $f : U \to \mathbb{C}$ be holomorphic and nonconstant. Let $z_0 \in U$ and let N be the order of the zero of $f(z) - f(z_0)$. Then there exists $\varepsilon, \delta > 0$ such that for each $w \in B_\delta(f(z_0)) \setminus \{f(z_0)\}$, the cardinality of $f^{-1}(w) \cap B_\varepsilon(z_0)$ is exactly N.*

Proof. In a small enough ball around z_0, we can write $f(z) = f(z_0) + (\varphi(z))^N$, where $\varphi(z) = (z - z_0)g(z)$ is holomorphic and $g(z_0) \neq 0$. In fact, φ is a biholomorphism from a neighborhood of z_0 to a neighborhood of $f(z_0)$, with $\varphi(z_0) = 0$. This fact is proved in the proof of the open mapping theorem, but we will reprove it here for completeness.

Note that
$$f(z) - f(z_0) = (z - z_0)^N h(z)$$
in a neighborhood of z_0, where h is holomorphic and $h(z_0) \neq 0$. Since h is continuous, there exists $\delta > 0$ such that $h(z)$ is nonzero on $B_\delta(z_0)$. So, there exists a holomorphic $g : B_\delta(z_0) \to \mathbb{C}$ such that $g^N = h$. Then,
$$f(z) - f(z_0) = [(z - z_0)g(z)]^N, \quad \forall z \in B_\delta(z_0).$$
Also, $\frac{\mathrm{d}}{\mathrm{d}z}[(z - z_0)g(z)] = g(z_0) \neq 0$, so the inverse function theorem implies that $\varphi(z) := (z - z_0)g(z)$ is invertible with differentiable inverse on a neighborhood of z_0. So φ is a biholomorphism. Also, $\gamma : w \mapsto w^n$ is N-to-1 on $V \setminus \{0\}$ for any neighborhood of 0. Thus composing, $f(z) = f(z_0) + (\varphi(z))^N$ is N-to-1 on a neighborhood of z_0 minus $\{z_0\}$. □

Corollary 5.10. *If $f : U \to \mathbb{C}$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$.*

Proof. Follows from the previous corollary with $N = 1$. □

Corollary 5.11 (Maximum Principle)**.** *Let U be an open connected domain. If $f : U \to \mathbb{C}$ is holomorphic and $|f|$ attains a local maximum in U, then f is constant.*

Proof. If $|f|$ attains a local maximum at some $z \in U$, then there is a ball around z such that its image under f does not contain an open ball around $f(z)$. Thus f is not an open mapping, and f is constant by the open mapping theorem. □

Remark 5.12. $|f|$ can attain a local minimum, but only if that minimum is 0. ☕

6 Harmonic Functions

Definition 6.1. Let $U \subset \mathbb{C}$ be open. A twice continuously differentiable function $u : U \to \mathbb{R}$ is **harmonic** if is has an everywhere vanishing Laplacian:

$$\Delta u(x + \mathrm{i}y) := \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \forall x + \mathrm{i}y \in U.$$

Lemma 6.2. *If $f : u + \mathrm{i}v : U \to \mathbb{C}$ is holomorphic, then u and v are both harmonic.*

Proof. Note that by the Cauchy-Riemann equations, we have

$$\partial_x^2 u = \partial_x \partial_y v = \partial_y \partial_x v = -\partial_y^2 u.$$

Similarly, $\partial_y^2 v = -\partial_x^2 v$. Thus the Laplacians of u and v vanish. □

Remark 6.3. This gives many examples of harmonic functions.

Example 6.4. $f(x + \mathrm{i}y) = e^{x+\mathrm{i}y}$ is holomorphic, so

$$u(x + \mathrm{i}y) = e^x \cos y, \quad v(x + \mathrm{i}y) = e^x \sin y$$

are harmonic.

Definition 6.5. If $u : U \to \mathbb{R}$ is harmonic and if $u + \mathrm{i}v$ is holomorphic, then we say v is a **harmonic conjugate** for u.

Theorem 6.6. *If U is a simply connected, a harmonic conjugate exists and is unique up to adding a constant.*

Proof. To see that the harmonic conjugate is unique up to adding a constant, suppose $v, \tilde{v}$ are both harmonic conjugates of u. Then $\mathrm{i}(v - \tilde{v}) = (u + \mathrm{i}v) - (u + \mathrm{i}\tilde{v})$ is holomorphic and purely imaginary. It is defined on a open connected set, so by the open mapping theorem, it is constant.

Next, we show existence. Let $f := \partial_x u - \mathrm{i}\partial_y u$. Since u is harmonic, so $\Delta u = 0$, i.e., $\partial_x^2 u = -\partial_y^2 u$. Denote $\tilde{U} := \partial_x u$ and $\tilde{V} := -\partial_y u$. Note that we have

$$\begin{aligned} \partial_x \tilde{U} &= \partial_x^2 u = -\partial_y^2 u = \partial_y \tilde{V}, \\ \partial_y \tilde{U} &= \partial_x \partial_y u = \partial_y \partial_x u = -\partial_x \tilde{V}. \end{aligned}$$

So f satisfies the Cauchy-Riemann equations and is holomorphic. Since U is simply connected, there exists a holomorphic $F : U \to \mathbb{C}$ such that $F' = f$. We claim that $u = \operatorname{Re} F$; then $v = \operatorname{Im} F$ is a harmonic conjugate for u. From the proof of Cauchy-Riemann equations, we have

$$\begin{aligned} F' &= \partial_x (\operatorname{Re} F) + \mathrm{i}\partial_y (\operatorname{Re} F) \\ &= f = \partial_x u - \mathrm{i}\partial_y u. \end{aligned}$$

So $\partial_x (\operatorname{Re} F - u) = 0$, so $\operatorname{Re} F - u$ depends only on y. We can similarly show $\partial_y (\operatorname{Re} F - u) = 0$, so $\operatorname{Re} F - u$ depends only on x. Thus $\operatorname{Re} F - u$ is constant, and by replacing F with $F -$ constant, we have $\operatorname{Re} F = u$. □

Remark 6.7. Note that we only used connectedness for uniqueness, but need simply connectedness for existence.

Example 6.8 (Finding harmonic conjecates). We can write down a holomorphic function with u as the real part (and taking the imaginary part), or solve using the Cauchy-Riemann equations.

Proposition 6.9. *Let $U, V \subset \mathbb{C}$ be open. Let $u : U \to \mathbb{R}$ be harmonic and $f : V \to U$ be holomorphic. Then $u \circ f$ is harmonic.*

Proof. Let $z \in U$. There exists $r > 0$ such that $B_r(z) \subset U$. Since $B_r(z)$ is simply connected, we can find harmonic conjugate v of u on $B_r(z)$. By definition of harmonic conjugate, $u + \mathrm{i}v$ is holomorphic on $B_r(z)$, and so $(u + \mathrm{i}v) \circ f$ is holomorphic. Its real part $u \circ f$ is harmonic. In particular, if $w \in f^{-1}(B_r(z))$, then $\Delta(u \circ f)w = 0$. □